

INT301 Bio-computation Week 4 Lab

Further Practice with Perceptrons

Demo:

Analyze the given program *PerceptronExample*, which consists of following parts:

- (1) Prepare *mydata* by generating 2 dimensional linearly separable data.

```
mydata = rand(500,2);  
% Separate the data into two classes  
acceptindex = abs(mydata(:,1)-mydata(:,2))>0.012;  
mydata = mydata(acceptindex,:); % data  
myclasses = mydata(:,1)>mydata(:,2); % labels  
[m n]=size(mydata);
```

You may check the data distribution using:

```
scatter(mydata(:,1),mydata(:,2))
```

- (2) Train the perceptron by calling the function *PerceptronTrn* with the prepared training data (x, y), which will return the connection weights, the bias, and the number of iteration;
- (3) Test the trained Perceptron model with the testing data (xt, yt), by calling another function *PerceptronTst*, which will return the testing error;
- (4) Display the two classes of data points with a separating line.

Exercise:

- (1) The Perceptron training function uses a learning rate 0.5 and a threshold 0.5. Change these two parameters, e.g. learning rate 0.1 and threshold 0, and observe the differences;
- (2) Revise the program to calculate the Root-Mean-Square (RMS) error for every input data points and display the error curve, i.e. RMS vs. iteration.